MBB: HOLLOW DESIGN ACCEPTED

```
cshell design accepted: mass saving 14.9% (V/V0 0.8510); K/K0 1.0200 / 1.0190, P/P0 1.0248 / 1.0245
(optimisation / shifted grid) ≥ 1, export mass-matched and watertight

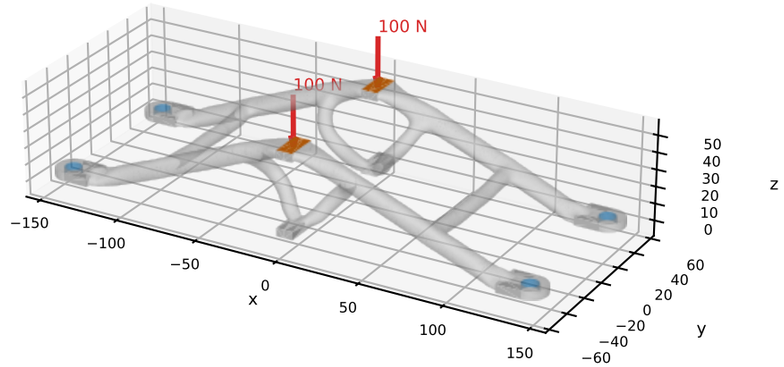
mass      target 75.43 g   final 75.42 g   error -1.9143e-04
optimiser grid  mass saving 0.149 (V/V0 0.851)  K/K0 1.02  P/P0 1.025  governing buckling  feasible True   (accept: K/K0 >=
    1, P/P0 >= 1)
shifted grid    mass saving 0.149 (V/V0 0.851)  K/K0 1.019  P/P0 1.025  governing buckling  feasible True   (accept: K/K0 >=
    1, P/P0 >= 1)

export    h 0.2 mm  faces 3755416  watertight True  volume 7.252e+04 mm3  material bodies 1  shells 3  enclosed voids 2
    escape holes 0
wall [mm] min 1.502  p05 1.51  median 2.26  max 25.84
closed_area_frac   0.2109
hollow_area_frac   0.6196

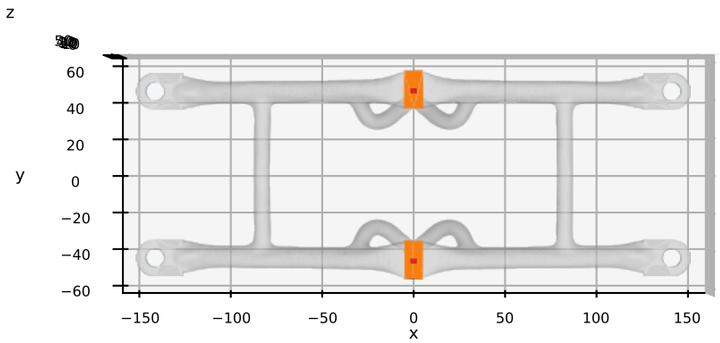
optimiser: {'method': 'mma', 'starts': ['solid'], 'max_iter': [60, 80, 40], 'move': 0.1, 'tol_x': 0.005, 'tol_f': 0.0001,
    'mass_tol': 0.0001, 'sliver_weight': [0.0, [0.0, 0.05], 0.1], 'seed': 0}
grid: {'phases': [{'h_e': 0.8, 'h_g': 0.2}, {'h_e': 0.4, 'h_g': 0.1}], 'export_h_mm': 0.05, 'rho_min': 1e-06,
    'tessellation_mm': 0.02, 'active_margin_mm': 0.5, 'geometry_device': 'auto', 'geometry_offload': True}
growth: {'enabled': True, 'phase': 0, 'tol': 0.001, 'patience': None, 'quantiles': [0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8,
    0.9, 1.0], 'step_tol': 0.0001, 'max_iters_per_step': 25, 'min_iters_per_step': 2, 'samples': 16}
objective: {'type': 'lightweight', 'lw_margin_stiffness': 1.02, 'lw_margin_capacity': 1.02, 'constraints': [],
    'constraint_scale': 0.01, 'feasibility_tol': 0.01, 'w_stiffness': 0.5, 'w_capacity': 0.5, 'min_gain': 0.02,
    'floor_stiffness': 0.85, 'floor_capacity': 0.85, 'floor_strength': None, 'floor_buckling': None, 'floors_as_constraints':
    True}
export cfg: {'h_mm': 0.2, 'device': 'gpu', 'closed_thr': 0.9, 'escape_holes': False}
```

MBB: supports (blue, all DOF fixed; 4 patches) and loads (orange faces; 2 red arrows, one per loaded patch at its area centroid, length and label = its share of the total 200 N). Faces read from bc_faces.npz (as analysed).

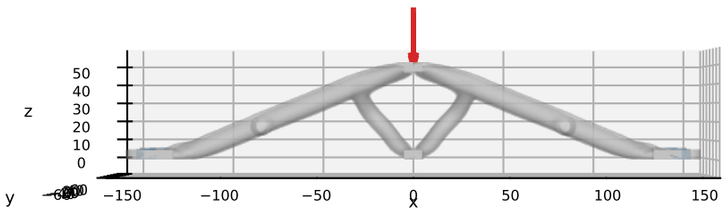
isometric



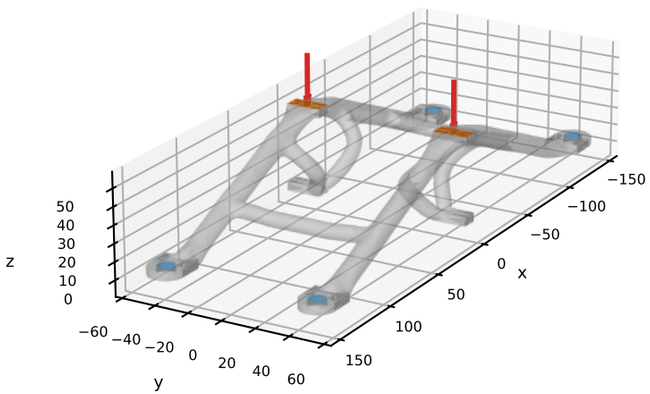
top (+z)



front (-y)

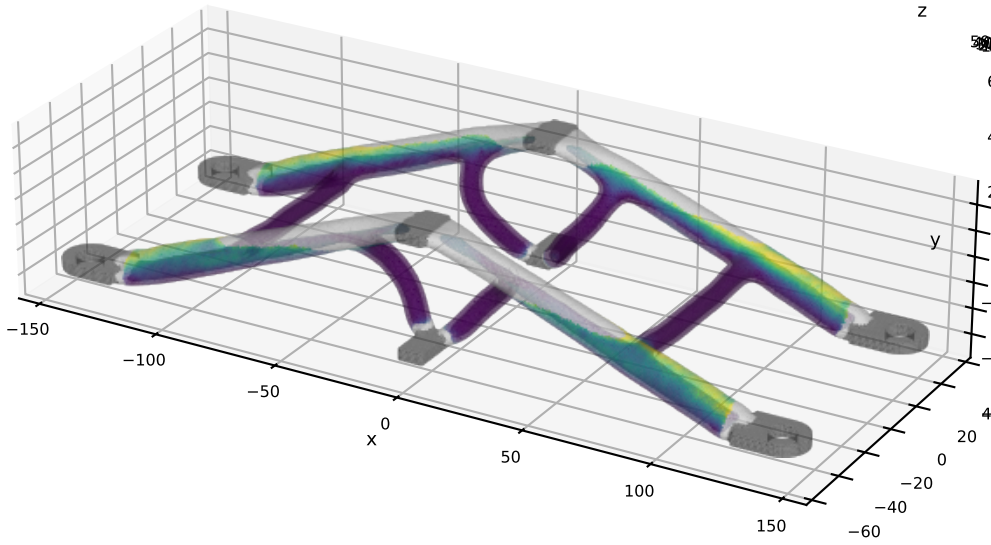


rear isometric

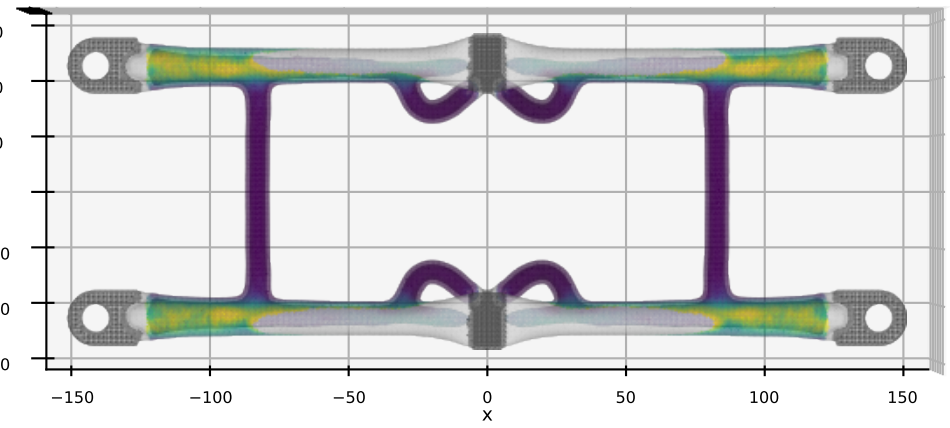


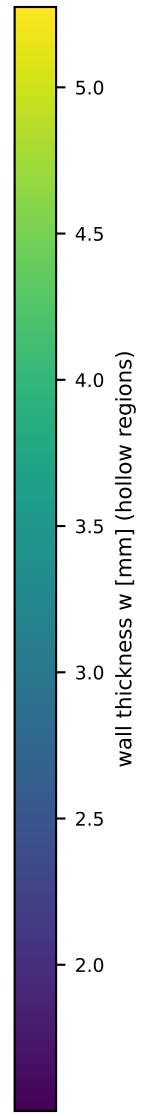
MBB: final design (STL 3,755,416 faces, shown decimated to 169,232; wall thickness from fields.vtp)

isometric

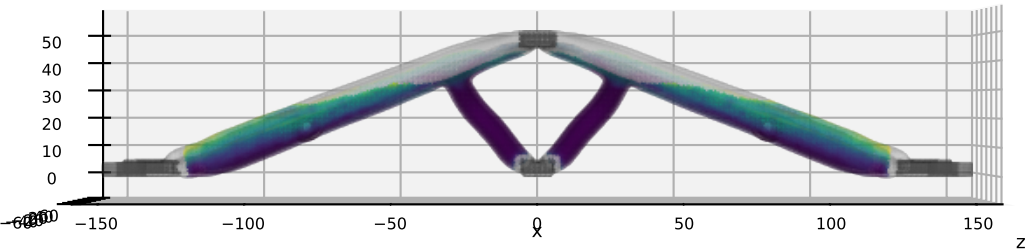


top (+z)

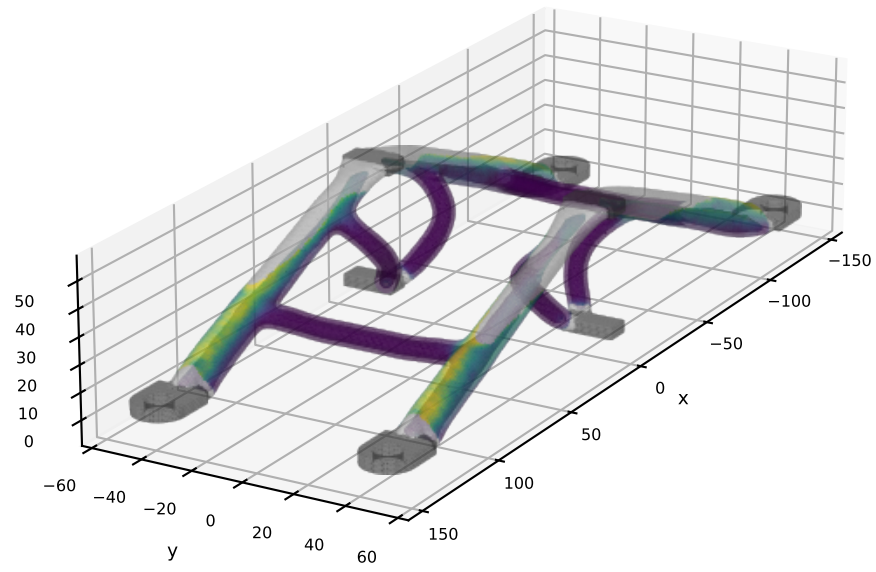




front (-y)

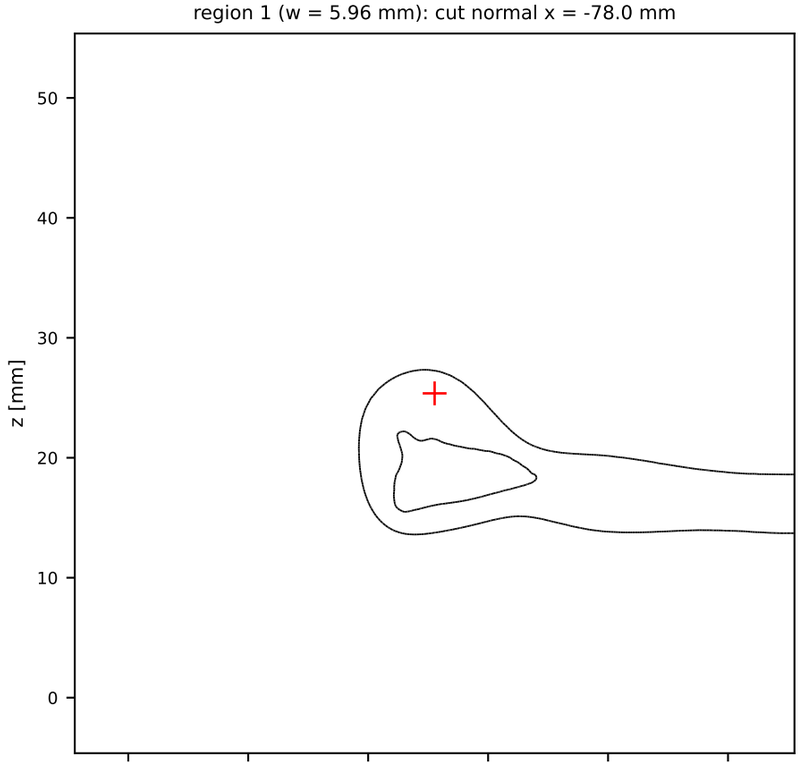


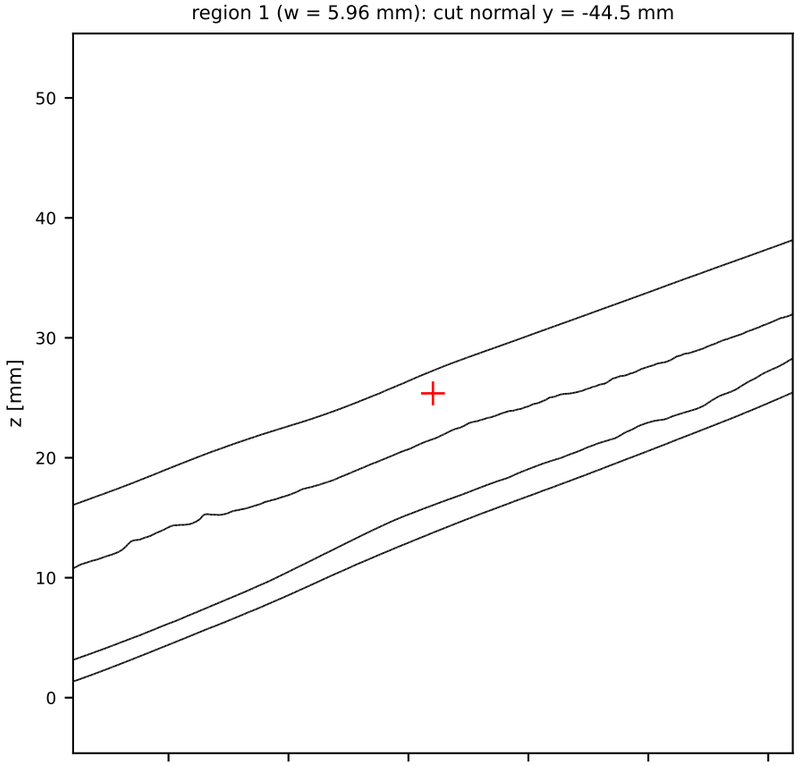
rear isometric

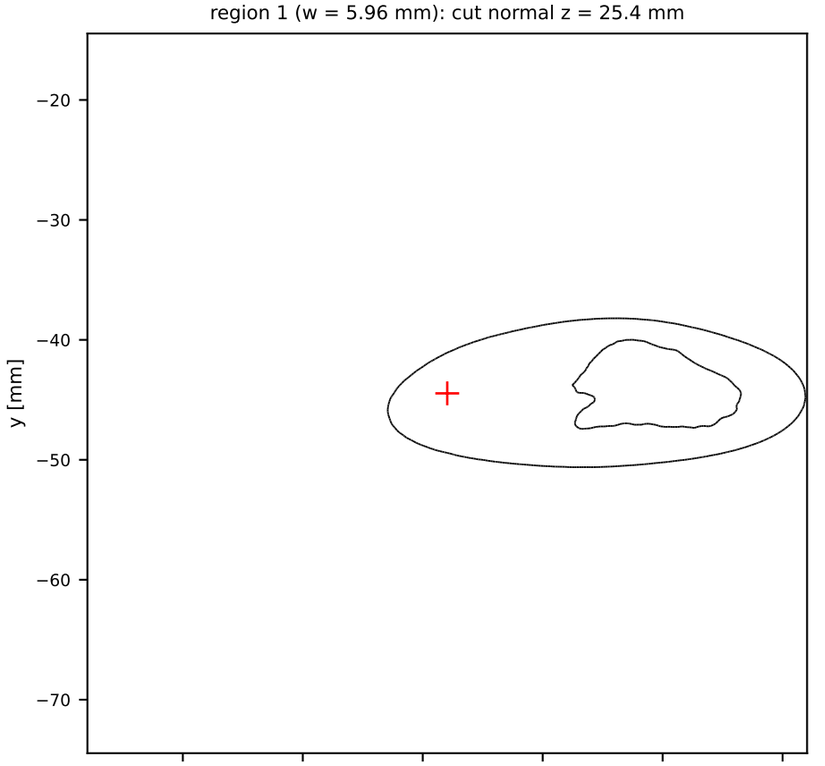


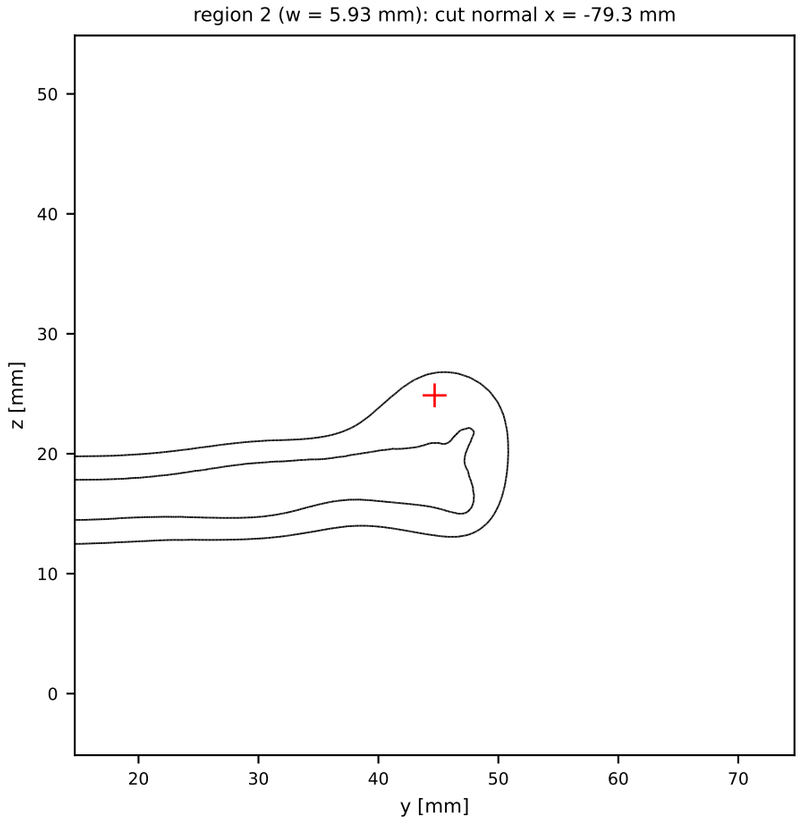
light grey = solid (closed) regions; dark grey = mounts; the colourbar covers the hollow regions only

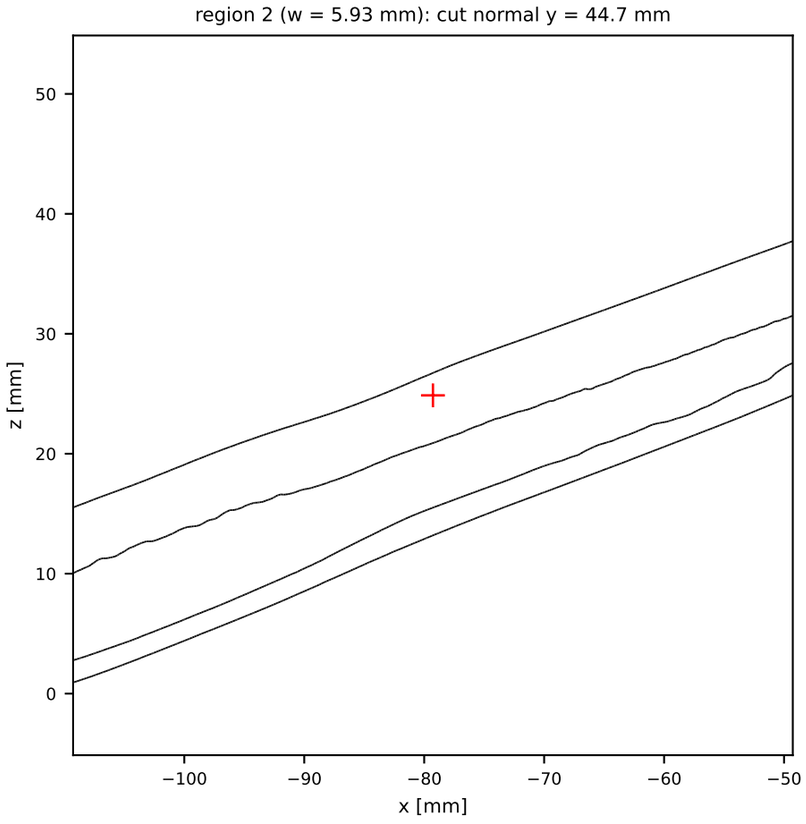
MBB: sections of the final design through the two thickest-wall regions (red +); +-30 mm window

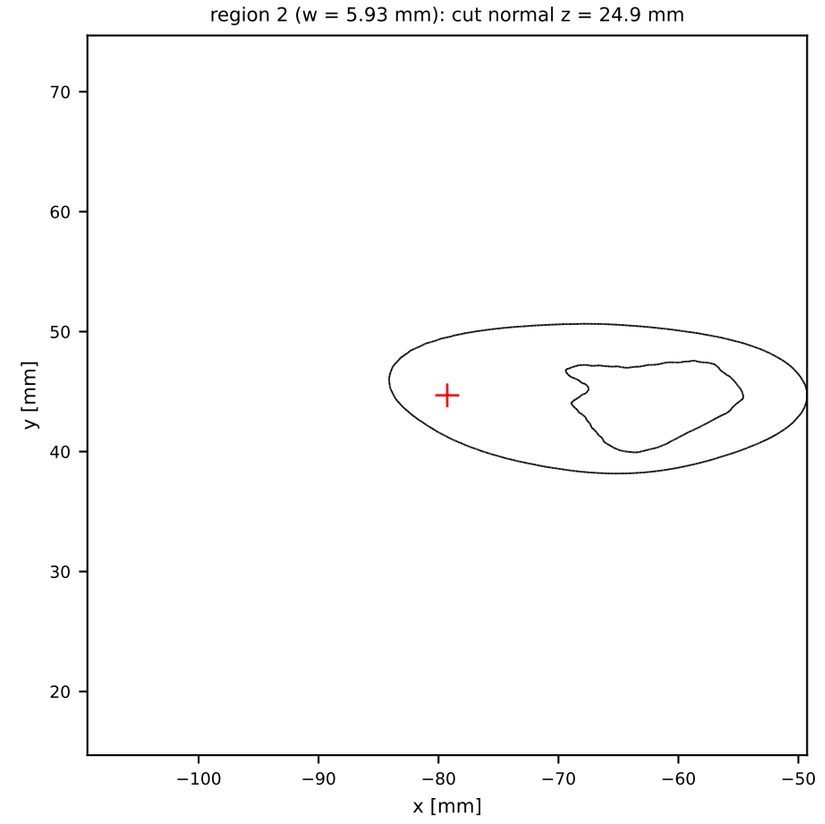






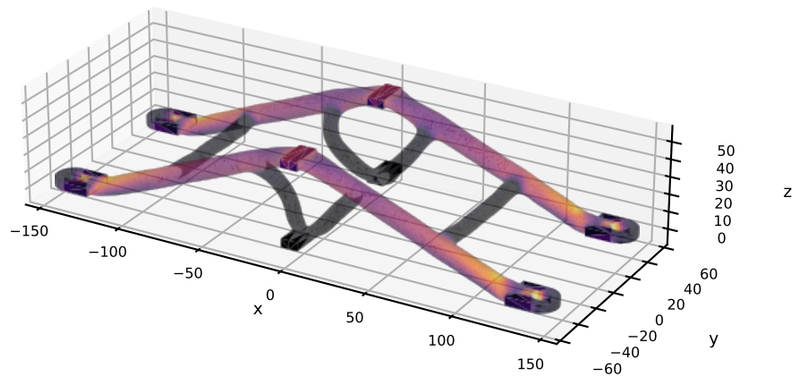




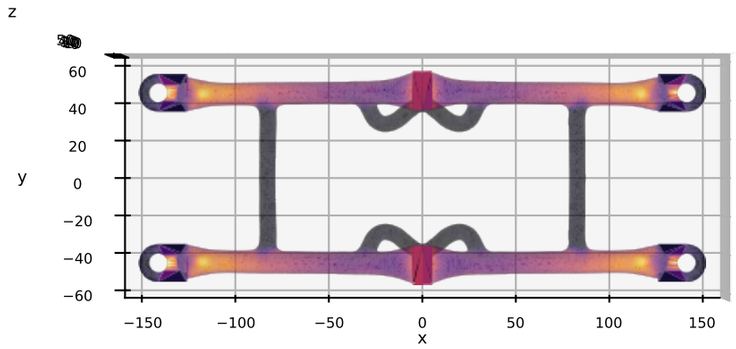


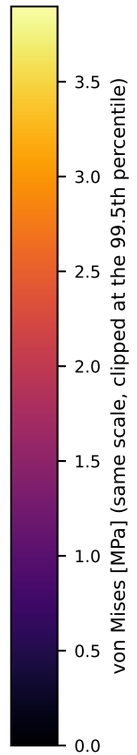
MBB: von Mises stress, original solid vs final design (peak solid 12.6, final 9.7 MPa; values from fields.vtp, nearest point)

solid: isometric

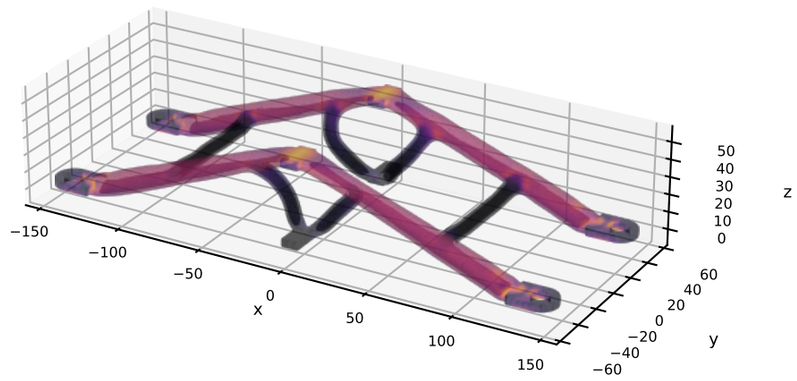


solid: top (+z)

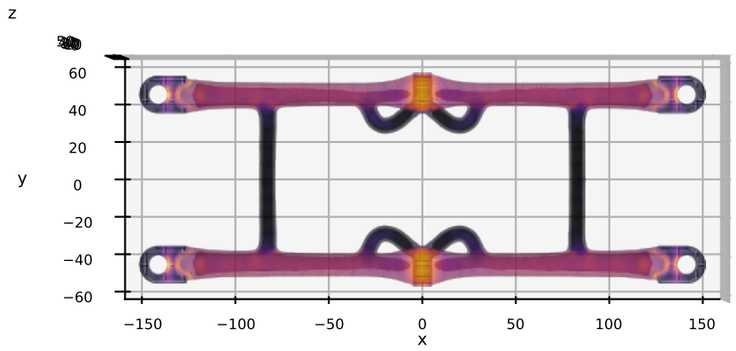


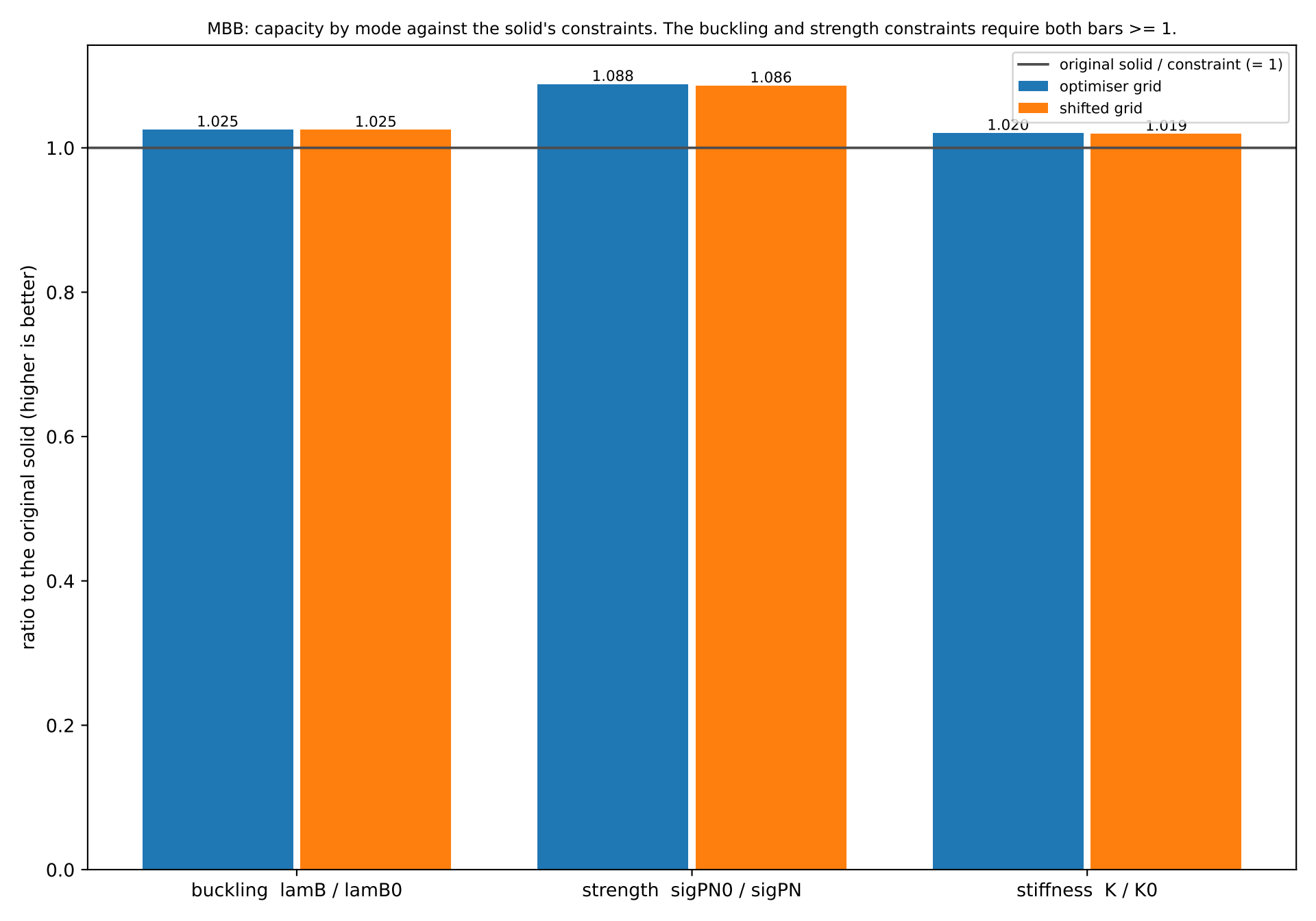


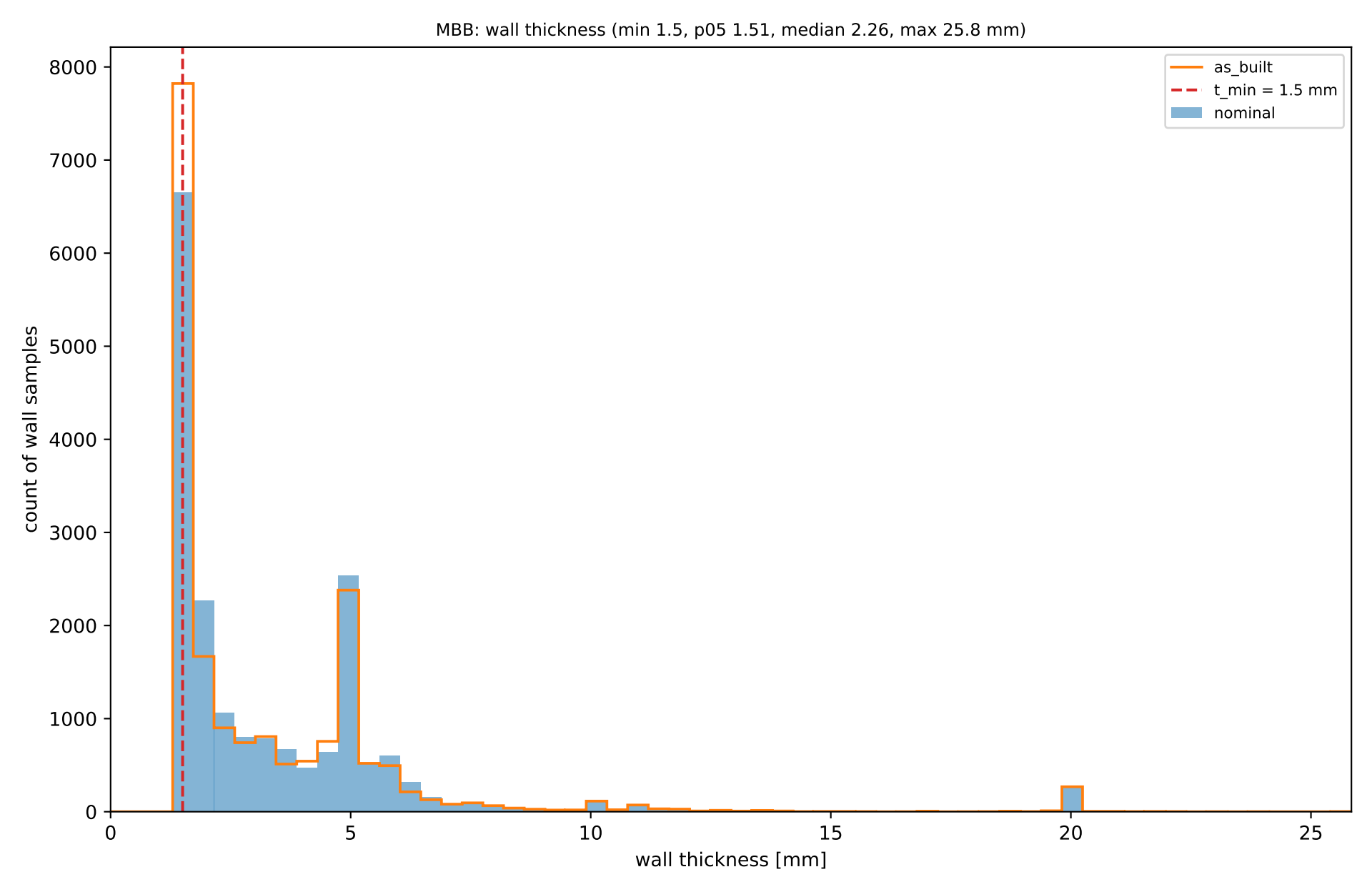
final design: isometric



final design: top (+z)

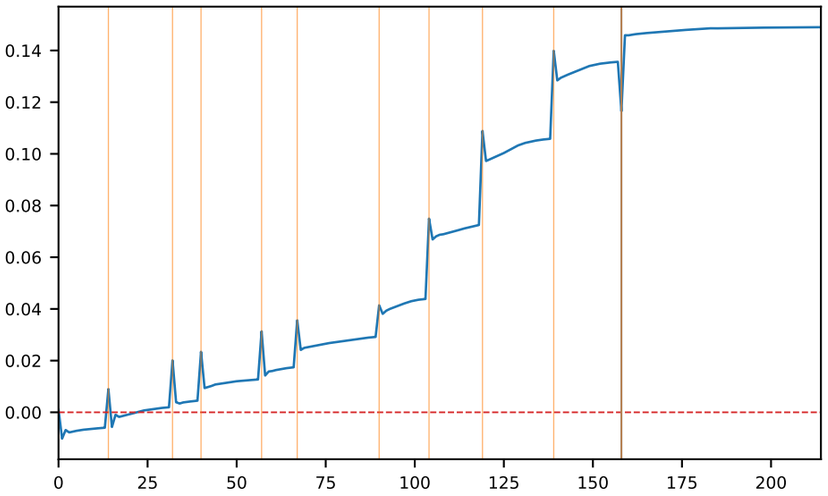



MBB: capacity by mode against the solid's constraints. The buckling and strength constraints require both bars >= 1.
original solid / constraint (= 1)
optimiser grid
shifted grid
ratio to the original solid (higher is better)
0.0
0.2
0.4
0.6
0.8
1.0
1.025
1.025
1.088
1.086
1.020
1.019
buckling lamB / lamB0
strength sigPN0 / sigPN
stiffness K / K0

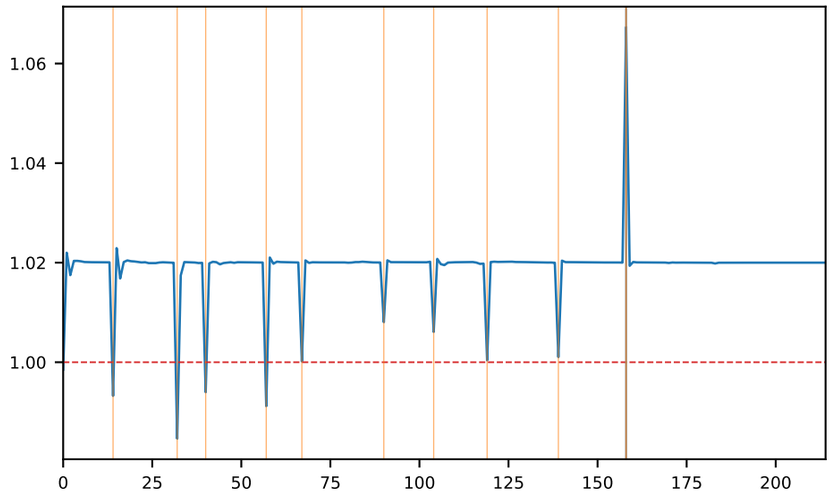

MBB: wall thickness (min 1.5, p05 1.51, median 2.26, max 25.8 mm)
as_built
t_min = 1.5 mm
nominal
count of wall samples
wall thickness [mm]
0
1000
2000
3000
4000
5000
6000
7000
8000
0
5
10
15
20
25

MBB: convergence history

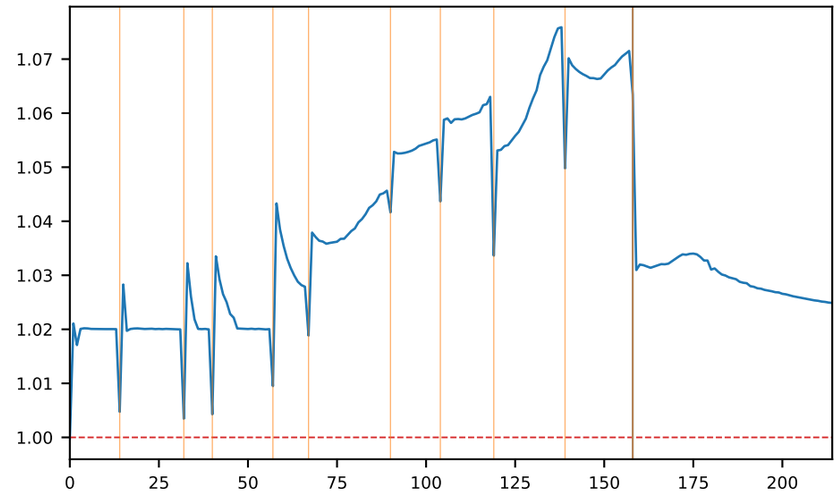
J = mass saving 1 - V/V0



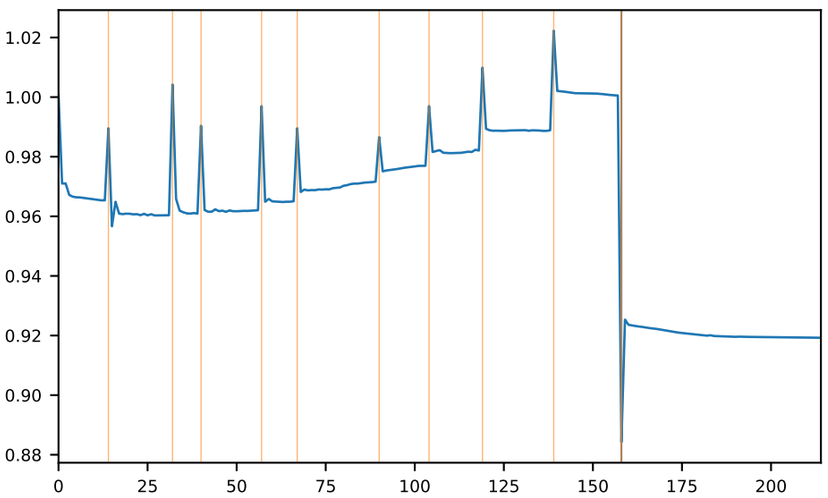
K / K0 (accept >= 1)



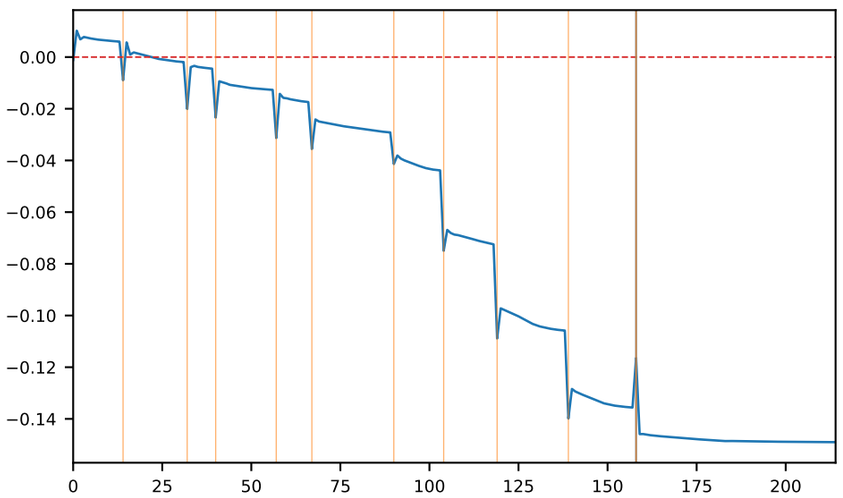
P / P0 (accept >= 1)



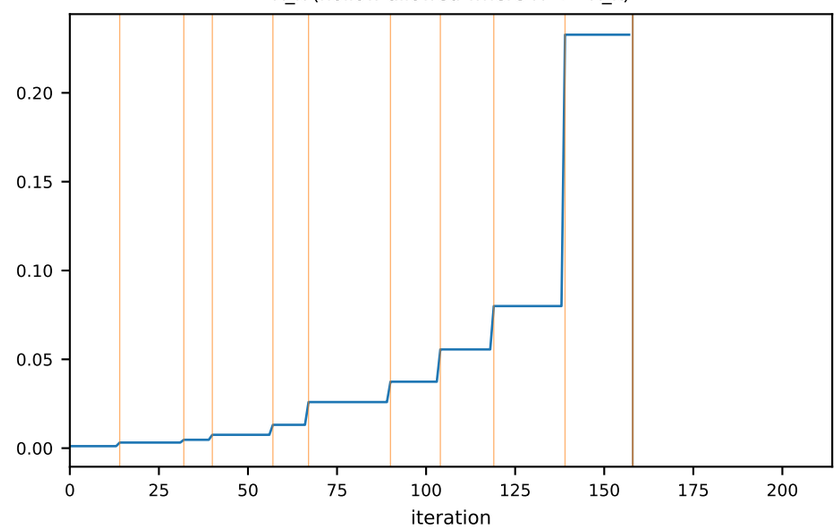
sigPN / sigPN0



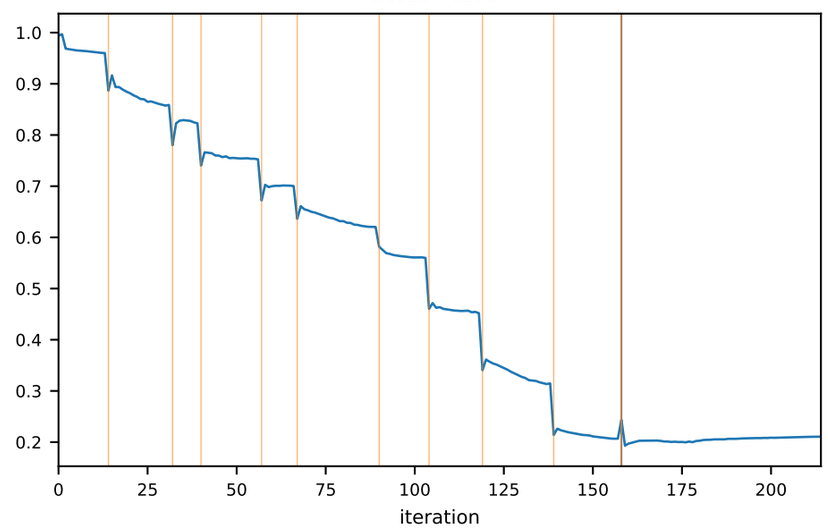
V/V0 - 1 (mass is free)



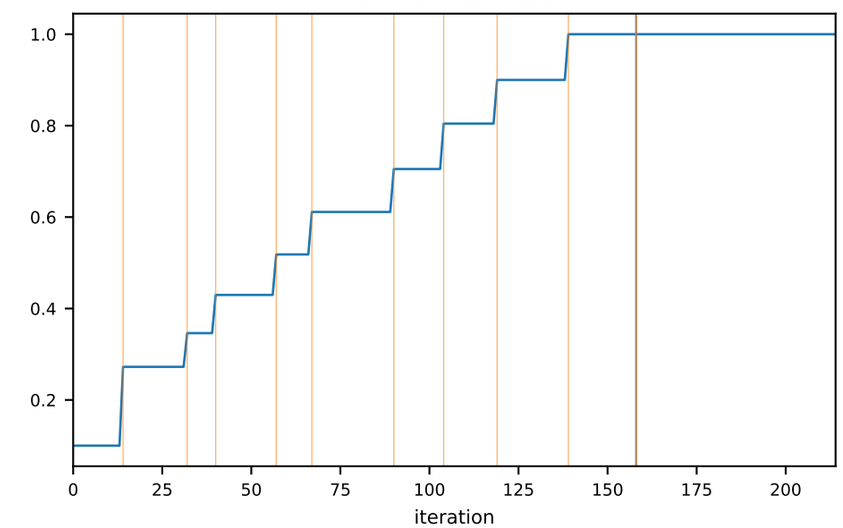
R_k (hollow allowed where R <= R_k)



closed fraction



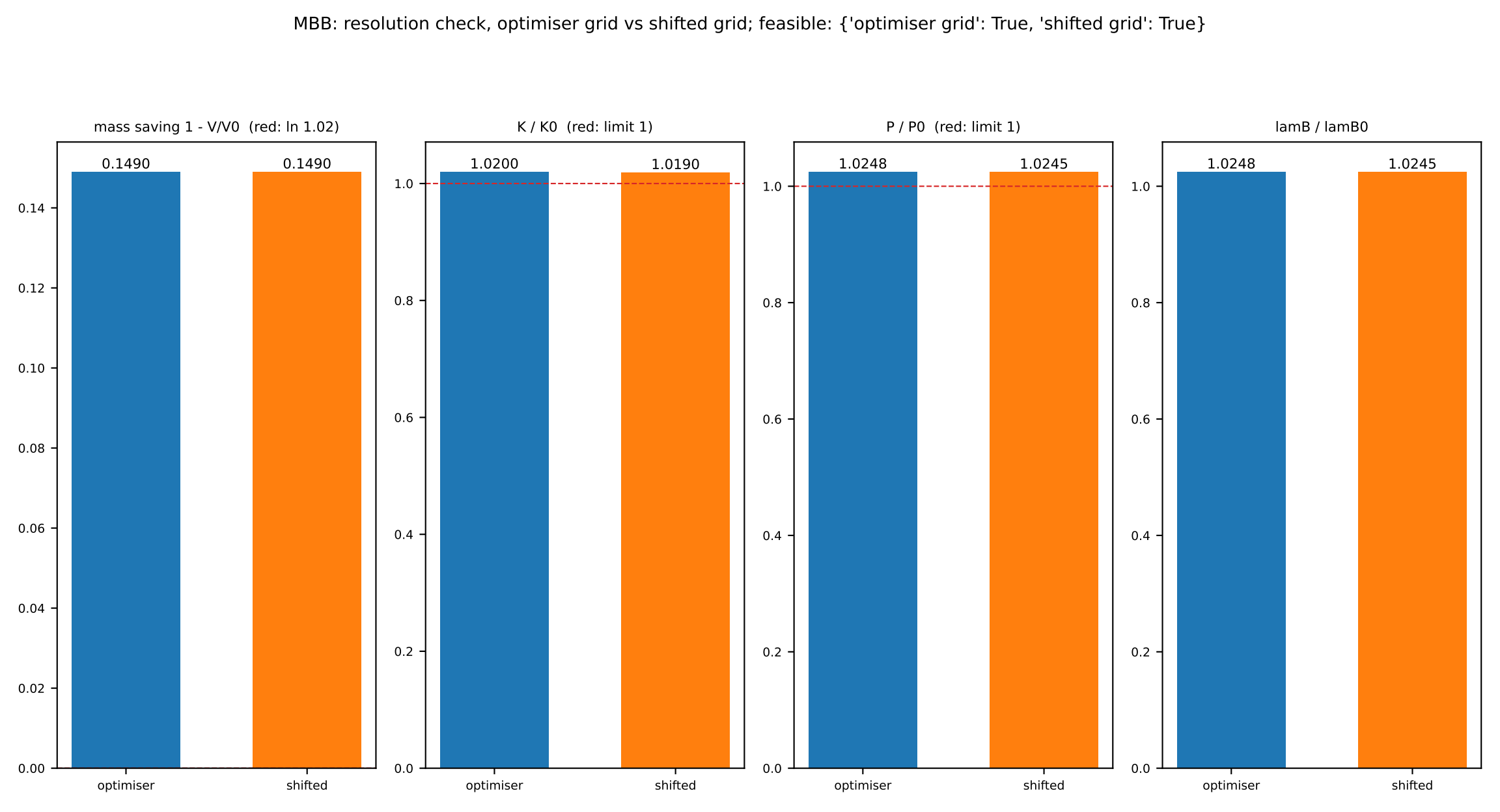
hollow-allowed fraction

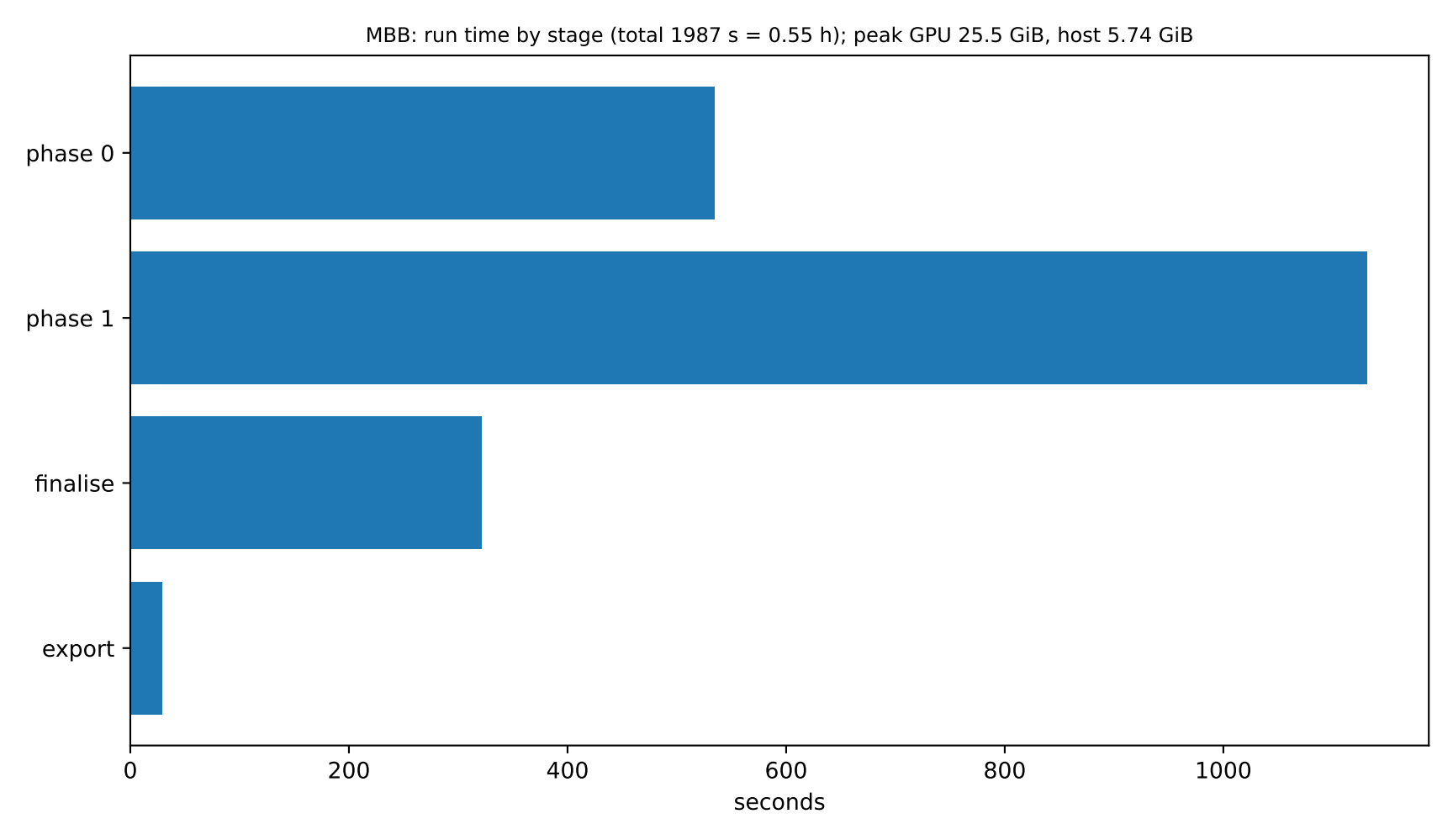


grey vertical: phase boundary
orange vertical: growth step (R_k rise)
iterations 215; growth steps 10
red dashed: constraint / acceptance level

```
enclosed voids: 2;  escape holes: 0;  voids without a hole: 2
  void 0: volume 11935 mm3, holes 0
  void 1: volume 11911 mm3, holes 0

Escape holes sit at the lowest-stress wall points of each void (outward direction in escape_holes.csv).
```


MBB: resolution check, optimiser grid vs shifted grid; feasible: {'optimiser grid': True, 'shifted grid': True}
mass saving 1 - V/V0 (red: ln 1.02)
0.1490
0.1490
K / K0 (red: limit 1)
1.0200
1.0190
P / P0 (red: limit 1)
1.0248
1.0245
lamB / lamB0
1.0248
1.0245
optimiser
shifted


MBB: run time by stage (total 1987 s = 0.55 h); peak GPU 25.5 GiB, host 5.74 GiB
phase 0
phase 1
finalise
export
0
200
400
600
800
1000
seconds